Bayesian Modeling for Defect Prediction

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Abstract

This paper explores the relationship between the number of days a product has been stored (X) and the number of defects observed on the product (Y). We employ a Bayesian approach to model this relationship, allowing us to rigorously and flexibly estimate uncertainties, especially when the sample size is small. Our objective is to determine whether there is a causal link between storage time and defects and to predict defect rates based on storage duration.

1 Problem and Initial Assumptions

Available Data

You have approximately $n \approx 100$ observations, each associated with:

- X_i: The number of days product i has been stored.
- Y_i: The number of defects observed on this product.

Objectives

- Determine whether there is a causal link between X (number of days) and Y (number of defects).
- Estimate the probability that a product belongs to a certain error range based on the number of days.

Necessary Assumptions

- **Assumption 1**: The data are independent and identically distributed (i.i.d.). This means that the pairs (X_i, Y_i) are generated independently according to the same joint distribution.
- **Assumption 2**: The number of defects Y can be modeled as a random variable dependent on X. We assume that $Y \mid X \sim \text{Poisson}(\lambda(X))$, where $\lambda(X)$ is a function of the number of days X.
- **Assumption 3**: We adopt a Bayesian approach, meaning we treat the model parameters as random variables with prior distributions.

2 Statistical Modeling

Probabilistic Model

We assume that the number of defects Y follows a Poisson distribution conditional on the number of days X:

$$Y \mid X \sim \text{Poisson}(\lambda(X)),$$

where $\lambda(X)$ is the average defect rate as a function of the number of days. A simple form for $\lambda(X)$ could be an exponential relationship:

$$\lambda(X) = \exp(\beta_0 + \beta_1 X),$$

where β_0 and β_1 are parameters to estimate.

Bayesian Approach

In a Bayesian framework, we define prior distributions for the parameters β_0 and β_1. For example:

$$\beta_0 \sim \mathcal{N}(0, \sigma_0^2), \quad \beta_1 \sim \mathcal{N}(0, \sigma_1^2),$$

where σ_0^2 and σ_1^2 are hyperparameters set according to prior knowledge.

The posterior distribution of the parameters β_0 and β_1 is given by Bayes' theorem:

$$p(\beta_0, \beta_1 \mid \text{data}) \propto p(\text{data} \mid \beta_0, \beta_1) \cdot p(\beta_0) \cdot p(\beta_1),$$

where $p(\text{data} \mid \beta_0, \beta_1)$ is the likelihood of the data under the model.

3 Likelihood of the Data

The likelihood of the data is obtained by combining the contributions of each observation (X_i, Y_i):

$$p(\text{data} \mid \beta_0, \beta_1) = \prod_{i=1}^{n} p(Y_i \mid X_i, \beta_0, \beta_1),$$

where $p(Y_i \mid X_i, \beta_0, \beta_1)$ is the probability density function of a Poisson distribution:

$$p(Y_i \mid X_i, \beta_0, \beta_1) = \frac{\lambda(X_i)^{Y_i} e^{-\lambda(X_i)}}{Y_i!},$$

with $\lambda(X_i) = \exp(\beta_0 + \beta_1 X_i)$.

Substituting $\lambda(X_i)$ into the likelihood expression, we obtain:

$$p(\text{data} \mid \beta_0, \beta_1) = \prod_{i=1}^{n} \frac{(\exp(\beta_0 + \beta_1 X_i))^{Y_i} \exp\left(-\exp(\beta_0 + \beta_1 X_i)\right)}{Y_i!}.$$

4 Bayesian Inference

To estimate the parameters β_0 and β_1, we use Bayesian inference methods:

Analytical Method

If possible, we compute the posterior distribution explicitly. However, in this case, the posterior distribution does not have a closed form due to the complexity of the likelihood. Therefore, we must resort to numerical methods.

Numerical Method

We use algorithms such as Gibbs sampling or Hamiltonian Monte Carlo (HMC) to approximate the posterior distribution. These algorithms generate samples from the posterior distribution, allowing us to estimate quantities of interest.

Once the posterior distributions of the parameters are estimated, we can:

- Calculate the probability of a causal link by examining whether $\beta_1 \neq 0$. For example, we can compute:

  $$P(\beta_1 \neq 0 \mid \text{data}) = 1 - P(\beta_1 = 0 \mid \text{data}),$$

  where $P(\beta_1 = 0 \mid \text{data})$ is estimated from the posterior samples.

- Predict the number of defects for a new product based on the number of days. The posterior predictive distribution is given by:

  $$p(Y_{\text{new}} \mid X_{\text{new}}, \text{data}) = \int p(Y_{\text{new}} \mid X_{\text{new}}, \beta_0, \beta_1)\, p(\beta_0, \beta_1 \mid \text{data})\, d\beta_0\, d\beta_1.$$

Code implementation

4.1 Step 1: Calculating the Probability of a Causal Link

The idea is to use the posterior samples of β_1 to estimate $P(\beta_1 \neq 0|\text{data})$. The following pseudo-code illustrates this process:

```
# Step 1: Obtain posterior samples for beta0 and beta1
posterior_samples = run_inference_algorithm(data)  # Returns posterior samples

# Extract beta1 samples
beta1_samples = posterior_samples[:, 1]  # Column corresponding to beta1

# Compute the probability that beta1     0
threshold = 1e-6  # Small threshold to consider beta1 as "non-zero"
prob_beta1_non_zero = sum(abs(sample) > threshold for sample in beta1_samples) / len(
    beta1_samples)

print("Probability␣that␣beta1␣   ␣0:", prob_beta1_non_zero)
```

Explanation: - `run_inference_algorithm(data)` represents the execution of a Bayesian algorithm (e.g., HMC via PyMC3 or Stan) that generates posterior samples for β_0 and β_1. - We extract the samples of β_1 and count how many of them are significantly different from zero (using a very small threshold, such as 10^{-6}). - The probability $P(\beta_1 \neq 0 \mid \text{data})$ is simply the proportion of samples for which $|\beta_1| >$ threshold.

4.2 Step 2: Predicting the Number of Defects for a New Product

To predict the number of defects Y_{new} for a new product with X_{new}, we use the posterior predictive distribution:

```
# Step 2: Predict the number of defects for a new product
def predict_defects(X_new, posterior_samples):
    predictions = []  # List to store predictions

    for sample in posterior_samples:
        beta0, beta1 = sample  # Extract  0  and  1  from the sample

        # Calculate   (X_new) = exp( 0  +  1  * X_new)
        lambda_new = exp(beta0 + beta1 * X_new)

        # Generate a sample of Y_new from a Poisson distribution
        Y_new_sample = sample_from_poisson(lambda_new)

        # Add this sample to the list of predictions
        predictions.append(Y_new_sample)

    return predictions

# Example usage
X_new = 10  # Number of days for the new product
predictions = predict_defects(X_new, posterior_samples)

# Calculate statistics on the predictions
mean_prediction = mean(predictions)
quantiles = calculate_quantiles(predictions, [0.05, 0.5, 0.95])  # Quantiles for
    credibility interval

print("Mean_prediction:", mean_prediction)
print("90%_credibility_interval:", quantiles)
```

Explanation: - For each posterior sample (β_0, β_1), we calculate $\lambda(X_{\text{new}}) = \exp(\beta_0 + \beta_1 X_{\text{new}})$. - From $\lambda(X_{\text{new}})$, we generate a sample of Y_{new} using a Poisson distribution. - We repeat this procedure for all posterior samples, resulting in a posterior predictive distribution for Y_{new}. - Finally, we compute statistics (mean, median, credibility intervals) on the predictions.

Expected Results

- **Probability of a causal link:** - If $P(\beta_1 \neq 0 \mid \text{data})$ is close to 1, this suggests a significant link between X and Y. - If it is close to 0, this indicates there is likely no link.
- **Prediction of the number of defects:** - The mean prediction provides a point estimate of the expected number of defects. - The credibility interval provides a plausible range for Y_{new}, accounting for uncertainty in the parameters.

5 Estimation of the Probability of Belonging to an Error Range

Suppose you define error ranges based on the number of defects, for example:

- Range 1: $Y < 5$,
- Range 2: $5 \leq Y < 10$,
- Range 3: $Y \geq 10$.

For a given number of days X, the probability of belonging to an error range can be calculated by integrating the posterior predictive distribution of Y:

$$P(Y \in \text{Range } k \mid X) = \int_{\text{values corresponding to range } k} p(Y \mid X, \beta_0, \beta_1)\, dY,$$

where $p(Y \mid X, \beta_0, \beta_1)$ is the probability density function of Y under the model.

In practice, this integral is often approximated numerically using posterior samples.

6 Exploration of Further Considerations

Model Validation

WAIC Criterion: The WAIC measures the quality of model fit while penalizing complexity. It is defined as:

$$\text{WAIC} = -2 \sum_{i=1}^{n} \log p(Y_i \mid X_i, \text{data}) + 2p_{\text{eff}},$$

where p_{eff} is the effective number of parameters.

Sensitivity to Priors

Test different prior distributions for β_0 and β_1, for example:

$$\beta_0 \sim \mathcal{N}(0, 10^2), \quad \beta_1 \sim \mathcal{N}(0, 10^2),$$

or more informative priors.

Alternative Models

Try nonlinear relationships between X and $\lambda(X)$, for example:

$$\lambda(X) = \exp(\beta_0 + \beta_1 X + \beta_2 X^2).$$

Causal Analysis

Use causal inference methods, such as structural models or Granger causality tests, to examine whether X influences Y.

Conclusion

By following these steps rigorously, you can determine whether there is a significant relationship between X and Y, and estimate the probability that a product belongs to a specific error range.